

Mu2e Genesis: Agentic Experiment Optimization

Workflow (i) — activities and near-term plan

detailed version — see the short deck for the talk

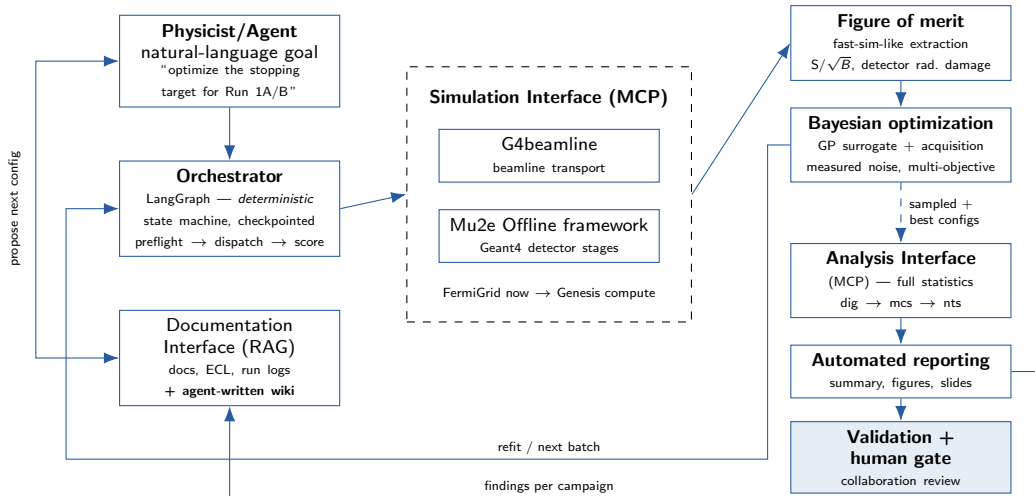
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Agentic Experiment Optimization — the loop

Prompt-to-configuration: simulate, analyze, score, decide — with a human gate.



Activities and owners (1/2)

Surrogate model optimization

(Yuri)

- ▶ GP surrogates over the configuration space; *measured* noise fed to the model; multi-objective acquisition over the true trade: **sensitivity vs. detector radiation damage**
- ▶ Needed: multi-fidelity, regime-aware objectives, transfer between scenarios, pre-registered predictions to grade the surrogate
- ▶ **Explore BO models for higher dimensionality:** TuRBO (rejected at $d=6$ — revisit at 10–12D), SAASBO (sparse axis-aligned; fits our few-dominant-knobs structure), input warping for corner non-stationarity

LangGraph orchestration

(Yuri / Ray / Rob)

- ▶ One graph run = one evaluation: propose → preflight → simulation → scoring; checkpointed, resumable, failure-classifying
- ▶ **Today the orchestrator is Claude:** LangGraph + Bayesian optimization carry the loop; the LLM agent turns the goal into campaigns, diagnoses failures, decides what runs next. Staying within Claude is a viable option — the alternative is a KE/MCP orchestrator over the same seams
- ▶ Needed: expose the loop through MCP; full tool/reasoning traces for reproducibility
- ▶ **Parameterize scenarios:** beam intensity, solenoid field scale, arbitrary geometry configuration — today only geometry knobs are variable, so a new scenario means hand-authoring a mode

Fast sim scoring for Run1A/B

(Michael)

- ▶ **The only scoring path in the loop today:** $S/\sqrt{B}$ and detector radiation damage extracted from the Geant4 output *without* full reconstruction — what makes search affordable
- ▶ Needed: **improve fast-sim performance; revisit Run 1B** (looked at before, needs redoing on the current stack) and new constraint scenarios; quantified agreement with the full analysis chain
- ▶ **Explore improvements and other fast-sim options:** beam optics upstream, detector response downstream

Activities and owners (2/2)

Full-sim validation (dig → mcs → nts)

(Sophie / James)

- ▶ **The high-fidelity path:** full chain at full statistics, autonomously callable — on selected configurations *and* on a **sampled fraction** (e.g. every 10th point), run **asynchronously** so it never blocks the next pick
- ▶ **Quantified relationship to the fast score**, and the judgment of whether it is good enough — sampling spreads paired points across the space, so the calibration comes for free
- ▶ Needed: per-configuration provenance; the **evidence bar and sign-off protocol** — replicates, systematics, background-regime scoping — before anything reaches collaboration review

Infrastructure — MCP, LLM, computing

(Ray / Rob)

- ▶ MCP servers, model endpoints, and compute allocations
- ▶ **Agent skills** — encoded procedures and gotchas; the proposal's "orchestration patterns", a packaged Phase-I artifact
- ▶ **Start on FermiGrid, transition to Genesis-provided compute** — the interfaces should make the backend a swap, not a rewrite
- ▶ Needed: token and CPU accounting per workflow — a stated project deliverable

Documentation interface

(Cole / Simon)

- ▶ Semantic search over documentation, ECL, shifter logs, run and configuration databases
- ▶ Needed: context assembly for the *optimizer* — which constraints are real vs. conventional

G4beamline

(Yuri)

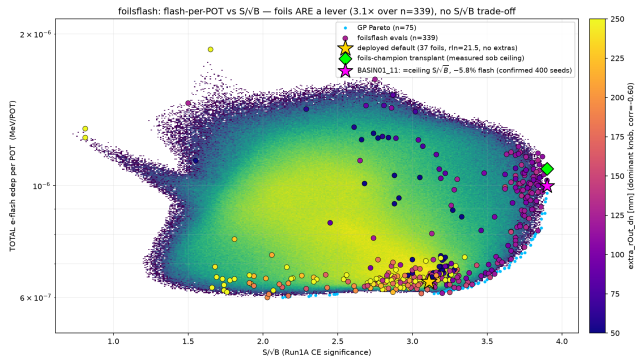
- ▶ G4beamline MCP server: autonomous configuration, job dispatch, output parsing — prototype demonstrated, **a proposal commitment** (App. 5)
- ▶ Needed: **integrate into the existing workflow** — the loop runs the Offline framework today; G4beamline plugs into the same seam, no new machinery

Automated reporting

(Rob / Yuri)

- ▶ A campaign drains → the loop **summarizes its own results**: regenerates the landscape and geometry figures, rebuilds the deck, and drafts the write-up
- ▶ **Write-back**: every campaign appends what it learned to the documentation interface — an agent-maintained wiki (Karpathy pattern, OKF v0.1). Running today at 125 pages; the write is per session, not yet triggered on drain
- ▶ Already partly real: figure generators are parameterized per line; today the assembly step is manual
- ▶ Needed: trigger on drain, **every number traceable to a leaderboard row**, and a human gate before anything circulates

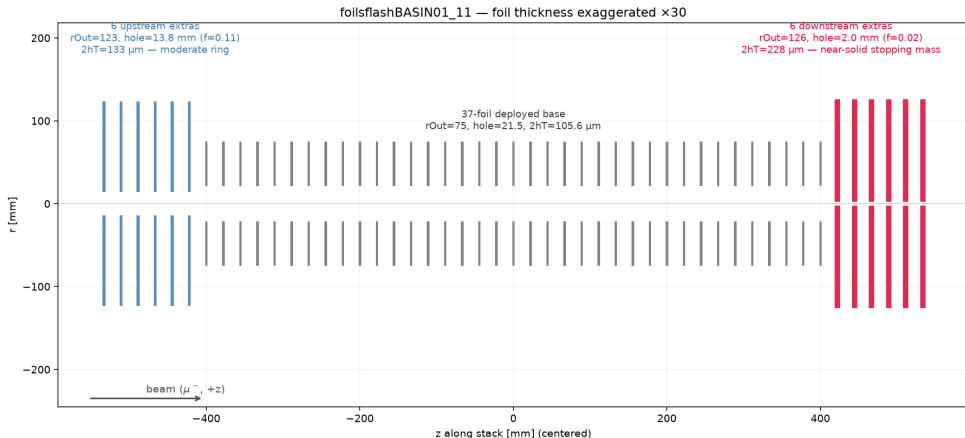
Status — the loop runs end-to-end



design	$S/\sqrt{B}$	rad. damage
deployed default	3.11	6.45×10^{-7}
ff11R00_07 (champion)	3.31	6.26×10^{-7}
ff14R01.00 (exploit)	3.84	8.14×10^{-7}
BASIN01.11 (confirmed)	3.90	1.00×10^{-6}

- ▶ **339** evaluations map a clean Pareto front
- ▶ Champion beats deployed on **both** axes: $+6\%$ $S/\sqrt{B}$ at -3% damage
- ▶ Best $S/\sqrt{B}$ **3.90 vs 3.11 deployed** = $+25\%$, confirmed reproducible (400-seed)

What the optimizer built — the $\max\text{-}S/\sqrt{B}$ target



Two routes to high $S/\sqrt{B}$: a thin **off-beam ring**, or thin **in-beam** material that degrades the beam to stop more muons. Here: 6 upstream extras (moderate ring) + 6 downstream (near-solid stopping mass) around the 37-foil deployed base.

Phase I milestones — where Workflow (i) stands

9 months from **1 Aug 2026**; go/no-go at month 6 (Jan 2027). **We are in month 1.**

window	dates	milestone (Workflow (i) parts)	status
Months 1–2	Aug–Sep 26	Deploy interfaces: Simulation, Analysis, Documentation End-to-end agentic chain: sim → analysis → FoM	pending done
Months 3–4	Oct–Nov 26	Exercise optimization on campaigns; collect metrics	ahead
Months 5–6	Dec–Jan 27	Compile go/no-go results; all traces logged	traces = gap
Months 7–9	Feb–Apr 27	Harden interfaces; package artifacts for Genesis	—

- ▶ One month in: the **science milestone of months 3–4 is already met**; interfaces are the open item
- ▶ Proposal §4 anticipates this: *“initial workflow prototyping can use simplified interfaces while production versions are improved”*

Decision gate metrics — Workflow (i), proposal §6

(a) *“Closed-loop sim–analysis–optimization (agent-orchestrated search with Bayesian optimization for local sampling) demonstrated on ≥ 2 distinct external-constraint scenarios (e.g. half beam intensity, 10% reduced fields), each initiated by a natural-language prompt and completed within one week (vs. months for the manual baseline), with full traces logged”*

- **Started.** “Optimize shape at the deployed foil pitch” → new mode spec, validated geometry, 20 evaluations in **7.3 h** (2026-08-04 23:34 → 08-05 06:51)
- **20 evaluations suffice:** a cold-start line hit sob 4.09 by eval 16 — matching a predecessor that took **414**. Saturation needs a second campaign (+0.00): a scenario answers **within a day** vs. a one-week bar
- Remaining: the two **external-constraint** scenarios — reduced / absent DS field, half beam intensity — and **formal trace logging**

(b) *“for a Run-1a scenario (no full cosmic ray veto coverage), starting from the collaboration’s existing manually-optimized baseline, agent-produced configuration achieves improved $R_{\mu e}$ sensitivity, with improvements documented for collaboration review”*

- **Demonstrated:** champion beats the deployed target on **both** axes (+6% $S/\sqrt{B}$, −3% damage); +**25%** $S/\sqrt{B}$ at the confirmed ceiling
- To finish: **prove it with full sims** (dig → mcs → nts) on the best configs
- Then: scope by background regime (part of the gain is Run-1a-specific); document for review

Backup

What “GP surrogate + multi-objective acquisition” means

GP surrogate — the map. Each real evaluation costs ~ 150 CPU-hours, so we cannot try many designs. The Gaussian Process looks at every design tried so far and draws a map of the whole space: for a design we have *not* tried it predicts the answer **and how unsure it is**. Near measured points the prediction is sharp; far away it is foggy. *The fog is the useful part.*

Acquisition — the rule for where to go next. Something has to choose which 10 designs to actually simulate next. The acquisition function trades off **exploit** (go where the map predicts good results) against **explore** (go where the fog is thick — we might be missing something).

Multi-objective — why there is no single “best”. We want two things at once: more signal, less detector damage. They fight each other, so there is no single winner — there is a **set** of best compromises, the **Pareto front** (the map on the status slide). Our acquisition function (qNEHVI) picks the designs expected to push that frontier *outward* the most, rather than chasing one number.